

Installation Instructions & Use Recommendations

FMS Accessories – CHP Motor Assembly

These instructions must be read carefully and thoroughly **before** beginning work. FMS Solutions, LLC accepts no liability for damage caused by failure to observe the installation instructions or use recommendations.

Installation of this accessory does not change the factory specified load limits for the motorcycle.

Only for units not factory-built to CHP spec.

Remove the dual siren speakers and install a new VO without the siren option 486, which will activate the horn function. Route the L & R harnesses up to the knee panel chambers on either side. Mount a civil horn to the motorcycle and connect using the horn wire on the RH side of the steering head.



Install the horn relay to the side panel using a cable tie.

Attach the 12x connector to the molded connector mount on the side panel and plug it into the dual siren harness.

Route the single wire of the “Y” harness to the civil horn and connect between the harness and horn for the “air horn tap”.

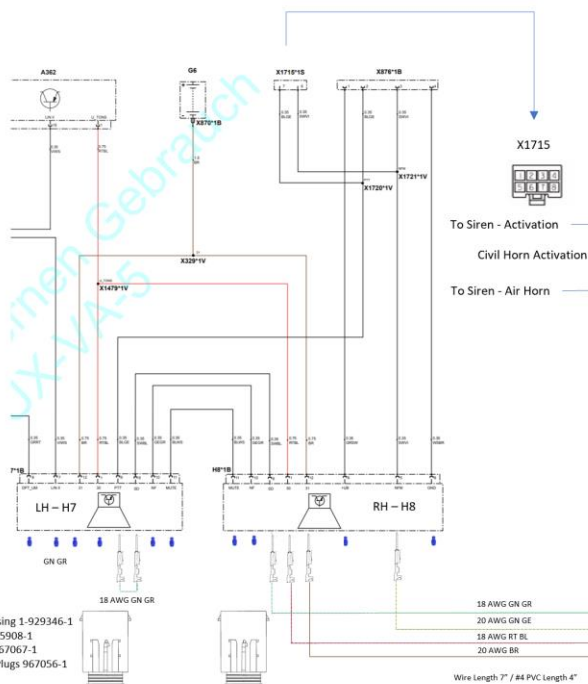


Install the loop plug to the LH siren harness lead and attach to the plug holder molded in the LH chamber.

Route your extra cables into the LH chamber.



You may use one of the cable tie “trees” to secure the wiring.



Horn / Siren Interface Wiring BMW PN 77 50 2 475 091 CHP Version R 1250 RT-P



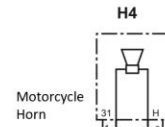
F1 = Siren Scroll
F2 = Aux Civil Horn (30 min. after ign. OFF)

- 1 F1 PTT Switch Signal (GMSP Pin 23 - 0.05A, momentary, 30 minutes after ign. off) SW GR
- 2 F2 PTT Switch Signal (GMSP Pin 24 - 0.05A, momentary, 30 minutes after ign. off) SW RT
- 3 Headset Interface Power (GMSP Pin 25 - 2A, 30 minutes after ign. off) SW VI
- 4 Headset - SW
- 5 Side Stand Extended Signal (direct connection to MC side stand switch in harness) GR GN
- 6 Microphone HF to Siren PA
- 7 PTT + to Siren PA (do not activate directly from F2 pin 2 or increased noise will result)
- 8 PTT 2 Switch Signal (GMSP Pin 21 - 0.05A, momentary, 30 minutes after ign. off) SW GN

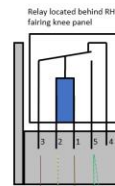
Color Coding:

BL = blue
BR = brown
GE = yellow
GN = green
GR = gray
RT = red
OR = orange
SW = black
VI = violet
WS = white
TR = transparent

1.3 = 16 AWG
.75 = 18 AWG
.5 = 20 AWG
.35 = 22 AWG



TE MQS Code B Socket
Housing 2-968790-1
Sockets 965906-1
Seals 967067-1



Relay Base
w/ Screw Hole
Wire Length 5"

Wire Length 24" / #8 PVC Length 22"
Wire Length 4.5" / #4 PVC 2.5"

From MC Harness

06/2019 FMS Solutions, LLC

Route radar cable to the front in the traditional way – photo shows approximate free length needed.



Route repeater antenna cable. Mount repeater antenna bracket to underside of LH pod by replacing the M5 bolt with the longer M5 fully-threaded bolt provided. Install rubber washer, bracket and KEPS nut and secure.

I found removing the LH color panel in front of the mirror and the radio speaker provided the room to route the cable up along side the emergency light cable.



Remove the M10 x 1.5 handlebar bolt. Clean the threads using a 10 x 1.5 tap to remove all old Loctite. Insert spacer. Install internal tooth lock washer on bolt. Mount stalker remote onto PTT bracket. Apply Loctite to bolt threads and secure hand-tight. Position bracket with a gap between the PTT bracket and the top of the clutch master cylinder. Pay attention to the fit of the tube as it passes between the clutch lever and high beam switch.

Torque to 40 Nm.

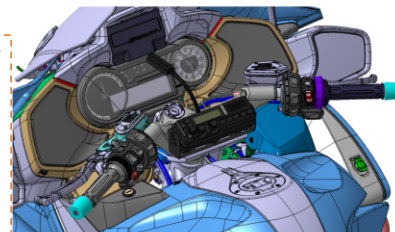
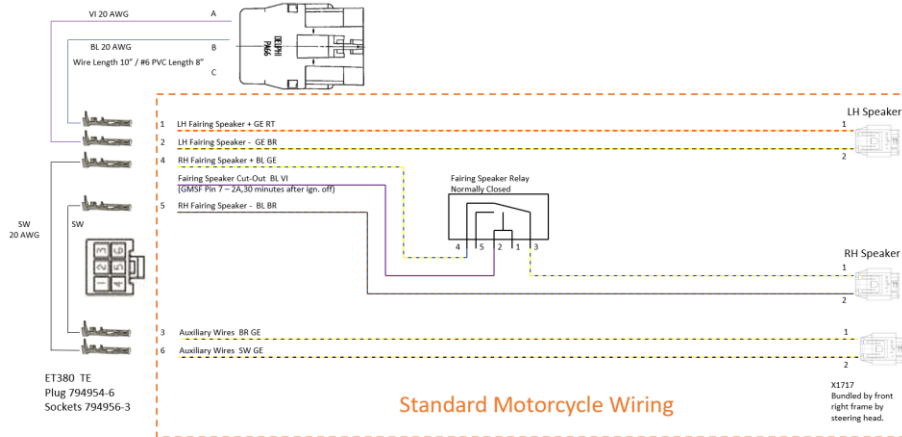


Radio Speaker Connection BMW PN 77 50 2 475 090 CHP Version R 1250 RT-P

BL = blue
BR = brown
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TR = transparent

Radio Box Speaker Adaptor
Plug Located Under Radio Box
Floor Plate for CHP Applications.

Apitiv 3P Pin
Housing 12020827
Pins 12089040
Seals 15324982
Plug 12010300



TE MQS Code A Pin
Housing 1-963658-4
Pins 965908-1
Seals 967067-1

Apitiv 2P Weatherpack Pin
Housing 12010973
Seals 15324982
Pins 1289040

Wire Length 3' / 1 1/4" #6 PVC

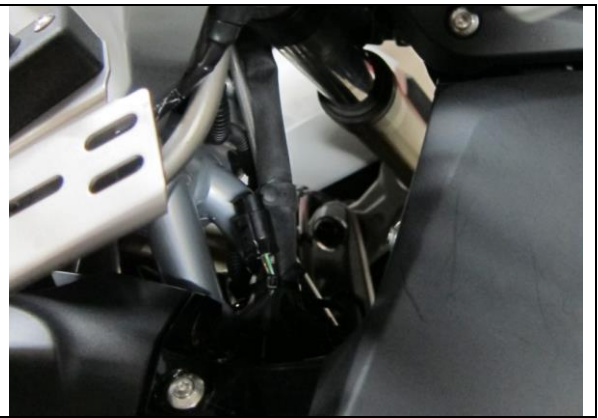
Radio Speaker Adaptor Plug by Steering Head

The radio speaker plug from the radio head connects to the radio speaker adaptor harness, which connects the speaker output from the radio head to the auxiliary harness wires on the left side of the steering head. The plug at the rear redirects the radio speaker signals back into the plug via two loop wires and sends it right back up to the RH speaker.

The radar signal from the white square amplifier box in the radio box connects to the other speaker input.

Route radio head cable up the LH handlebar cable pathway and loop around under support bracket – secure in radio head. You will need to stuff some extra unused Weatherpack plugs in the triangle area under the bracket.

I like the microphone bracket for the clip, but it was not in the original bid. CHP supplies the radio head bracket with a microphone clip.



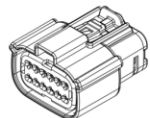
3955 Siren



Color Coding:

BL = blue
BR = brown
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Siren Wiring Interface BMW PN 77 50 2 475 089 CHP Version R 1250 RT-P



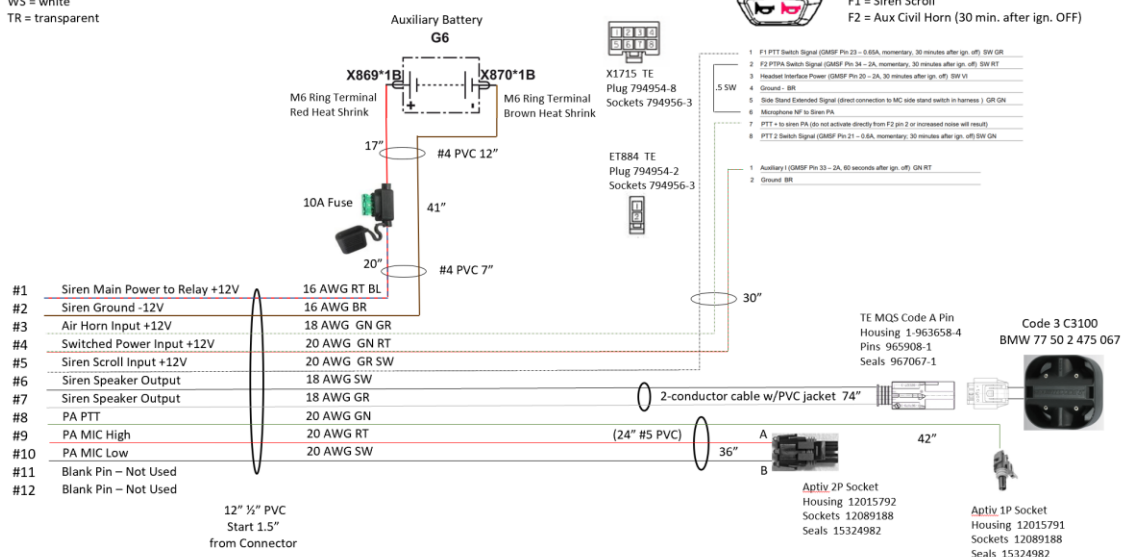
Molex 0334721206



Molex 0330122002



Molex 0343450001



The siren amplifier mounts to the inside face of the RH saddlebag between the bag mount and the frame. Drop the bag mounting bolts through the bracket on the back side and locate / drill the M5. Install clip nut on bracket.

Mount amplifier to bracket using supplied M6 self-clinching studs and serrated washer-head nuts.

Mount assembly to the bag securing with the one M5 washer-head Torx bolt. Mount bag to the motorcycle.



The siren harness main plug runs from the amplifier up through the gap between the side panel. You may need to remove the screw supporting the bottom of the panel to separate the parts enough to run the harness up from the outside to the inside.

Leave a gentle loop of cable to ensure that the plug on the amplifier is not stressed – the heavy PVC sheath should protect the cable from abrasion.



Route siren harness along the inside of the frame locating the waterproof fuse holder near the radio box support mount. The main power cable for the radio system will run along the outside of the frame.

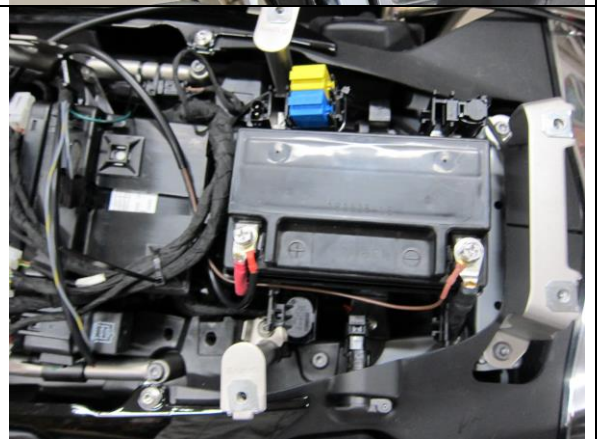
Connect the + and – to the battery ... there is no parasitic load as there is a main relay inside the siren that is triggered by the ACC1 plug connection.



Remove the plug bracket. Lift the GMSF up and route the main cable under the GMSF. Secure the relay block with cable ties to the battery case. Plug the radio power plug into the battery case molded holder.

Attach the various cables from the radio power connection kit, radio speaker connections and siren harness while the radio box plate is away.

Connect the raw end of the braided ground strap to the battery negative terminal (remove lock washer) to be routed to the top side of the radio box base plate.



Mount siren speaker to upper bracket using special pan-head bolt provided with speaker – securing through holes provided for driver. Secure lower mount using same bolts. Install front radar antenna using the M5 self-clinching bolts and serrated washer-head nuts.

Now, install speaker onto LH protection bar and secure using bolts provided. Cable tie siren cable to support as shown.



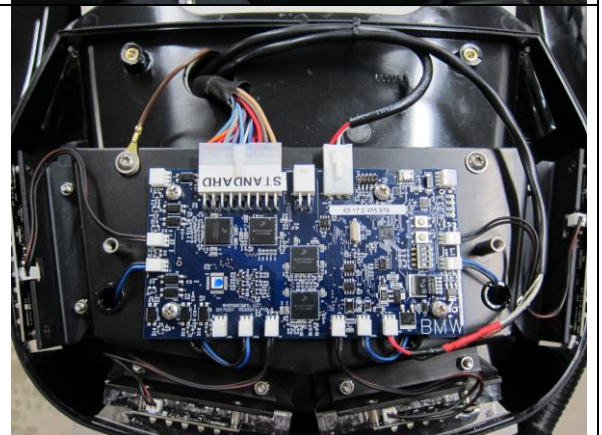
The ground strap from the battery connects to the bolt securing the radio mounting. A second ground cable attaches here and runs up the Code 3 light cables to the rear pod where it grounds the heat sink.



A special CHP flasher board has been created which is restricted only to Title 13 and CHP approved activities (photo not accurate – board PN 77 50 2 475 069).

Note ground wire connection to heat sink.

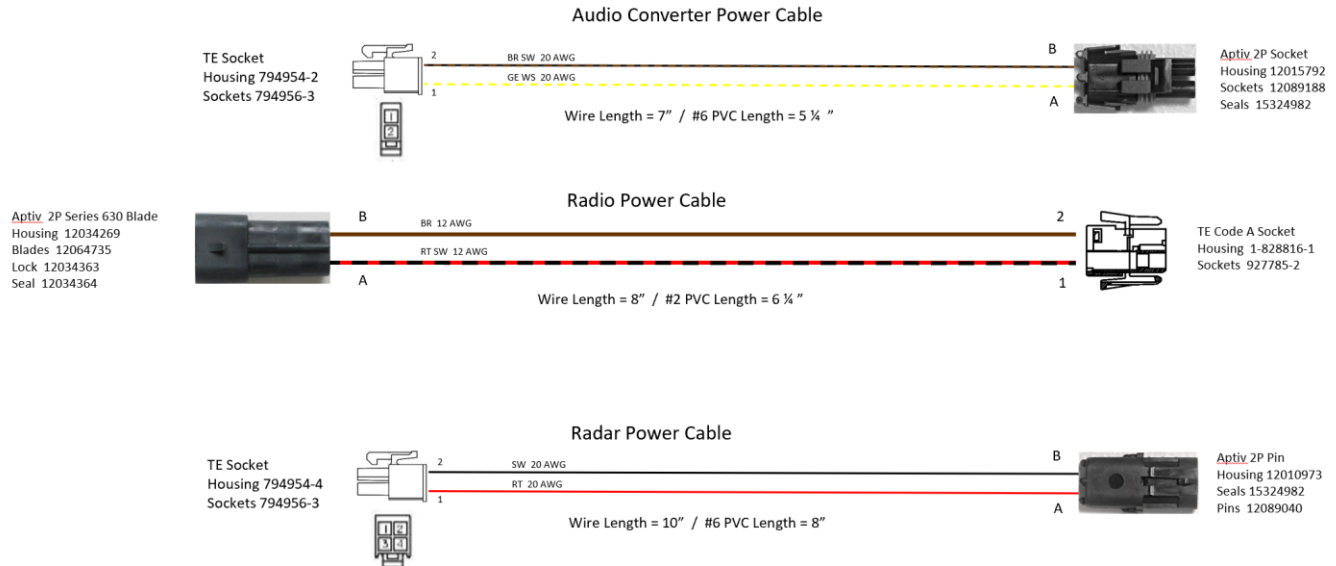
Note ID light harness only uses single plug.



Radio Power Connector Kit

BMW PN 77 50 2 475 092

CHP Version R 1250 RT-P

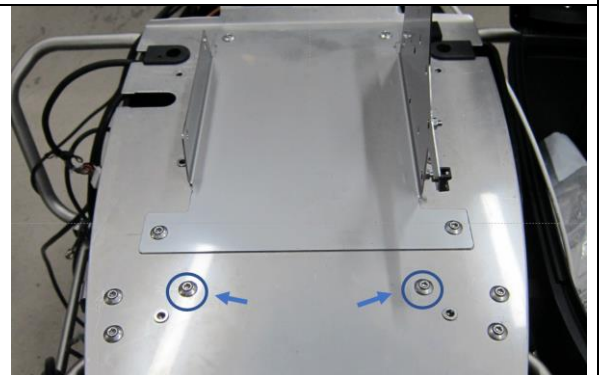


Connect the cables above to the outputs on the MC harness under the floor of the radio box. Each output should be long enough to enable connect of the adaptor harness to the CHP component.

Audio converter Power Cable to ACC III (RH Side)

Connect the 18-wire harness under the floor of the radio box and bring it up through the forward wire port.

Upon reassembly, replace these two M5 x 16 bolts securing the skirt trim with bolts provided in radio bracket kit. The new bolts are shorter on top and will allow the radio to settle down more, enabling everything to stack up and let the lid close.



Wiring is segregated to various ports to reduce the likelihood of interference between the systems.

The left front port has only wiring from the lighting system (18-wire cable, 3-wire duplex light cable and ID light cable) as well as the thick cable from the front radar display.

All other radio & repeater cable are routed through the second cable port.



The right side cable port is for the radio power cable, audio converter power cable (thin long black cable) and the radio antenna cable. As always, do your best to keep the antenna cable away from the other cables ... never cable tie them to other wires.

Antenna Cable Length = 26" above grommet.



Other Cable Lengths as measured above grommet in radio box:

Black HSA Cable = 8"

Front Radar Antenna Cable = 13.5"

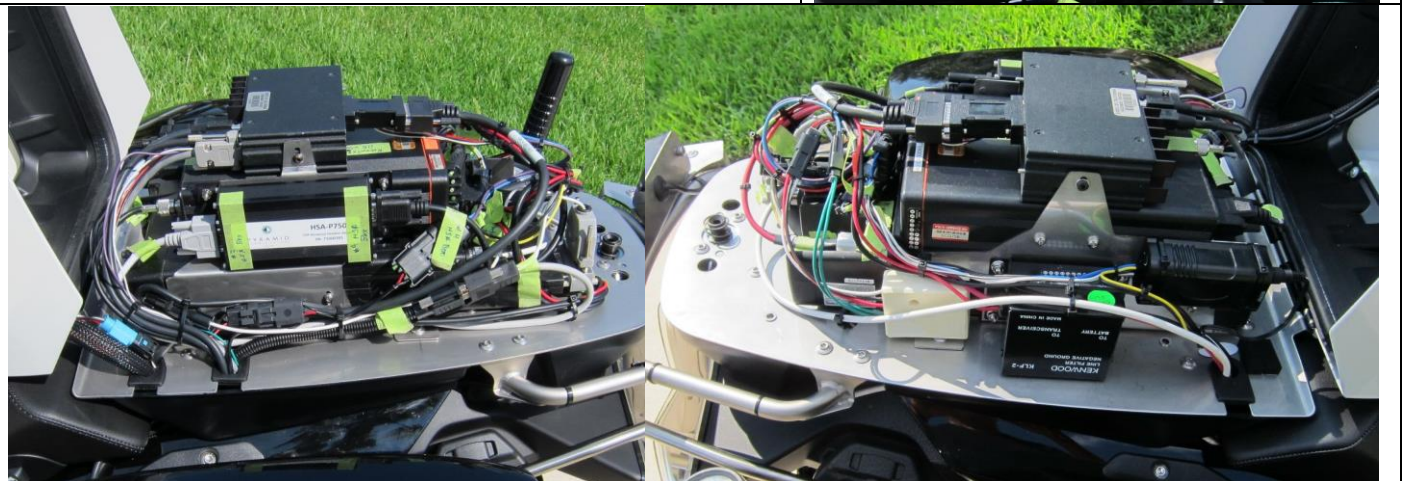
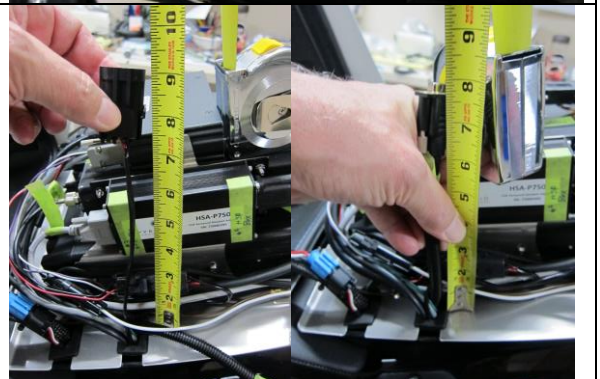
Rear Radar Antenna Cable = 13"

Stalker Power Cable = 13"

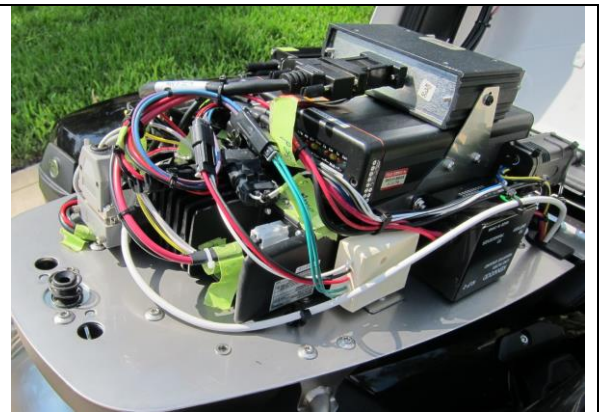
Repeater Antenna Cable = 9.5"

Radar Speaker Wire = 9"

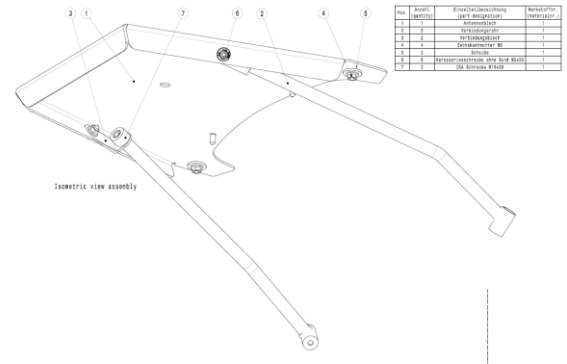
Radar Counter Unit (from display head) = 21"



Use on of the provided cable tie “tree” mounts to keep the antenna cable separated. Note the 90 degree D25 adaptor plug on the left and the ivory box on the right is the radar audio amplifier box, which connects to the Weatherpack 3x connector with only one of the three locations plugged.



When mounting the radio antenna mount, remember to install the warning label provided in the Code 3 kit warning that the antenna mounting base is not a lift point.



Mount the ID lights and rear radar antenna as shown at right. A special cable has been created for the ID lights to minimize space, with only one plug at the flasher board.

Route both cables along the saddlebag mount and up to the radio box.



Mount front radar antenna as noted.

Note that a warning is lasered into the speaker mount stating that no added weight is allowed on the left side. Warn any CHP customer that added weight is not allowed should they decide to get creative and mount something on the left side.

